

Duplicate grant line items in the 990 grants extracts

To: GivingTuesday Data Commons team **From:** Vibrant Data Labs (Zein Tawil) **Date:** July 31, 2026

Extracts reviewed: Schedule I grants (“Grants to Domestic Organizations”) batch 2026_06_25; 990-PF grants (“Current Grants” / Part XV) batch 2026_06_16

Summary

While reconciling grant totals against filers’ own declared totals, we found that both grants extracts contain large volumes of duplicated line items — the same grant appearing two (sometimes four) times. Three distinct mechanisms are involved, and they are additive:

1. **Every filing version is included.** When a filer amends a return, the extracts carry the grant line items of *both* the original and the amended filing for the same tax year, with no column that marks which filing is current.
2. **Whole blocks are emitted twice.** Some filer-years’ entire grant block appears twice verbatim. On the 990-PF side this affects thousands of filers in the most recent batches and is *not related to amendments* — it looks like a processing defect introduced around the taxyear-2024 batch.
3. **One provenance error.** In at least one case (Fidelity Charitable’s 2021 Schedule I) the original and amended filings’ blocks are both present but both stamped with the *original* filing’s `url` and `files` hashes, so they cannot be separated using the file metadata.

Across the Schedule I extract this inflates 2020+ grant dollars by roughly **\$36B (~5%)**; on the 990-PF side the recent-batch doubling alone adds roughly **\$7B**, concentrated in taxyear 2024. Per-filer, the distortion is much larger — an affected filer’s totals read at exactly 2× reality.

We have working per-filer detectors and complete lists of affected filer-years for every category, and are happy to share them.

How we found it

We load the extracts into Postgres without modification and routinely reconcile the summed line items for a filer-year against the total the filer itself declared on the return (990 Part IX line 1; 990-PF Part I line 25, column (d)). Two observations started the investigation:

- **Fidelity Charitable, tax year 2021:** the Schedule I line items sum to ~\$20B in cash grants against a declared total of \$11.39B — almost exactly double.
- After removing duplicate rows, dozens of large filer-years reconciled to their declared totals **to the dollar**, which rules out coincidence and filer error.

We first verified the duplication is present in the source extracts, not introduced by our loading: our loaded row counts match the extract files exactly (e.g. 8,991,013 Schedule I rows in batch 2026_06_25), and the duplicate rows are visible in the raw files themselves.

For the Fidelity case we went one step further and pulled the two underlying IRS e-file XMLs — 202321309349304807_public.xml (original) and 202430459349302913_public.xml (amended, AmendedReturnInd set). The extract’s two physical blocks match the original and amended XML exactly on (recipient EIN, cash amount) multisets — proving the extract contains both filings’ line items while labeling both with the original filing’s url/filesha256.

What we found, in detail

A. Multiple filing versions per filer-year (both extracts)

When both an original and an amended return exist for a filer-year, the extracts include the grant line items of each, usually distinguishable only because each version carries its own url.

- Schedule I: **3,084 filer-years** with 2+ urls.
- 990-PF grants: **10,772 filer-years** with 2+ urls (largest: EIN 46-2626883, tax year 2023 — 363,676 rows).

Any consumer that sums line items by (EIN, tax year) — the natural grouping — double-counts these filers. Nothing in the data flags that a version supersedes another.

B. Verbatim block doubling under a single filing

Some filer-years’ entire block appears exactly twice under one url.

- **Schedule I: 634 filer-years (~\$18B of raw line-item dollars).** Nearly all of the larger cases (377 of 378 with ≥40 rows) have an amended filing on record whose content is identical to the original — consistent with the amendment’s identical block being emitted under the original’s label.
- **990-PF: 6,277 filer-years, ~\$7.0B of inflation — and only 5 of the 6,277 have any amended filing on record.** This is a different mechanism from (A): the doubling is concentrated in the newest batches — **5,506 of the filer-years are tax year 2024** (\$6.59B), 545 are 2025, 190 are 2023, and years ≤2022 have single-digit counts. That pattern points to a defect in recent batch processing rather than anything filers did.

The decisive evidence that these are spurious: half the block’s sum equals the filer’s declared total *to the dollar*. Examples:

Filer EIN	Tax year	Line items sum to	Filer declared
23-7093598	2024	\$712,181,958	\$356,090,979 (= exactly half)
52-1512330	2024	\$374,001,262	\$187,000,631 (= exactly half)
47-2107200	2024	\$1,397,879,334	\$698,939,667 (= exactly half)

The last case (Sergey Brin Family Foundation) also shows that the doubling copies the whole block: line items the filer legitimately listed twice appear four times.

C. Provenance mislabeling (confirmed once, may exist elsewhere)

Fidelity Charitable (EIN 11-0303001), tax year 2021: both the original filing's block (64,828 rows) and the amended filing's block (65,494 rows) are present, and **both carry the original filing's url and filesha256**. Unlike categories A and B, this cannot be repaired from the extract alone — the version labels are wrong at the source. This is the case we verified against the raw IRS XMLs.

What would fix it upstream

In order of value to consumers:

1. **Emit one filing version per filer-year (the current one), or add an explicit version/amended indicator column** so consumers can filter deterministically. Today the only signal is the `url`, and nothing marks which url is current.
2. **Investigate the 990-PF batch doubling (category B)**. Its concentration in the 2024-batch, its independence from amendments, and the exact-half reconciliation should make it findable in the extract-generation step. This is the highest-dollar recent issue and presumably affects every downstream consumer of the PF grants extract.
3. **Fix version labeling (category C)** so each block carries the `url/filesha256` of the filing it actually came from.

We can provide: complete affected filer-year lists for every category (CSV), the SQL detectors we use, and before/after totals for verification. We've also implemented interim dedup rules on our side and are glad to share the methodology and edge cases (e.g. distinguishing spurious doubles from filers that legitimately repeat identical grants).

Appendix: compact detector (Schedule I)

Flags filer-years with multiple filing versions, verbatim doubles, or sums far above the declared total. Column names as in the extract.

```

WITH g AS (
  SELECT filerein, taxyear, COUNT(*) AS n_rows,
         COUNT(DISTINCT md5(ROW(rteinorecipi, rtnbbnline11, retaamofcagr,
                                rtaoncassist, rectabaddcit, rectabaddsta,
                                retapuofgrra)::text)) AS n_distinct,
         COUNT(DISTINCT url) AS n_urls,
         SUM(CASE WHEN retaamofcagr ~ '^-[0-9]+(\.[0-9]+)?$'
                   THEN retaamofcagr::numeric ELSE 0 END) AS cash
  FROM grants_to_domestic_organizations GROUP BY 1, 2
), b AS (
  SELECT filerein, taxyear, COUNT(DISTINCT filesha256) AS n_filings,
         MAX(CASE WHEN graallpaitot ~ '^-[0-9]+(\.[0-9]+)?$'
                   THEN graallpaitot::numeric END) AS declared
  FROM basic_fields GROUP BY 1, 2
)
SELECT g.*, b.n_filings, b.declared
FROM g LEFT JOIN b USING (filerein, taxyear)
WHERE g.n_urls > 1 -- multiple versions
      OR (g.n_rows = 2 * g.n_distinct AND g.n_rows >= 40) -- verbatim double
      OR (g.n_urls = 1 AND b.n_filings >= 2) -- mislabeled
version
      AND g.cash > 1.5 * b.declared);

```

The 990-PF detector is analogous (declared total = Part I line 25 column (d)); for the category-B doubling, the discriminating test is that **half** the block sum reconciles with the declared total better than the full sum does.